



Document No.	QSL-CAL-PROC-001
Revision / Date	1.0 — 26 June 2026
Approved By	Technical Manager

CALIBRATION PROCEDURE

Non-Automatic Weighing Instruments (NAWI) — per EURAMET Calibration Guide No. 18

1. Scope & Application

This procedure applies to the calibration of electronic non-automatic weighing instruments (NAWI), including platform scales, bench scales, counting scales, and weighbridges, across single-range, multi-range, and multi-interval instrument types. It is based on EURAMET Calibration Guide No. 18, "Guidelines on the Calibration of Non-Automatic Weighing Instruments", and is consistent with OIML R76.

2. Reference Standards & Traceability

STANDARD	DESCRIPTION	TRACEABILITY
Class M1 / F1 Weight Set	1 mg – 20 kg, selected per test load and instrument capacity	National metrology institute, ≤ 36 months
Reference Thermo-Hygrometer	Ambient temperature and relative humidity monitoring	Accredited lab, ≤ 12 months
Barometer	Ambient pressure for air buoyancy correction	Accredited lab, ≤ 12 months

3. Environmental Conditions

PARAMETER	REQUIREMENT
Location	Vibration-free, draught-free surface; instrument levelled per manufacturer specification
Temperature Stability	Recorded at start and end of test; instrument allowed to stabilise to ambient before testing
Air Currents	No direct airflow (HVAC vents, open windows, fans) onto the weighing pan

4. Procedure

- Pre-Test Inspection.** Inspect the instrument for visible damage, verify level, and confirm the display zeroes correctly with an empty pan after warm-up per manufacturer specification.
- Repeatability Test.** Apply a single test load of approximately 0.5-Max (single range) at least 5 times under identical conditions, recording the indication after each application. Re-zero between applications if required.
Per cg-18: load applied ≥ 5 times (≥ 3 times where the test load is ≥ 100 kg); load should, where possible, consist of one single body.
- Eccentricity Test.** Apply a test load of approximately Max/3 (or higher) sequentially at the centre, front-left, rear-left, rear-right, and front-right positions of the load receptor, zeroing between each application and recording the indication.
Per cg-18: omit only where construction-related restrictions make eccentric loading impossible.
- Error of Indication.** Apply a minimum of 5 test loads distributed across the weighing range (typically zero/Min, 0.25-Max, 0.5-Max, 0.75-Max, Max), increasing by steps with unloading between each step, and record the indication at each point. Use 11 test loads for finer resolution where required by the client or critical application.
Per cg-18: "Zero" is a valid test load; deviations of up to 10% of Max (5 test loads) or 5% of Max (11 test loads) from a targeted load value are acceptable.

5. **Multi-Range / Multi-Interval Instruments.** Repeat steps 2–4 separately for each partial weighing range or range to be calibrated, per the test-point allocation in Section 5 of cg-18.
6. **Minimum Weight Determination.** Using the standard deviation from the repeatability test (and, where applicable, an additional repeatability measurement at small load), calculate the minimum weight at which the instrument meets a specified relative weighing tolerance.

5. Uncertainty & Calculation

Standard deviation from the repeatability test, the eccentricity deviation, the resolution of the instrument, the calibration uncertainty of the reference weights, and the air buoyancy correction are combined per the GUM method to determine the expanded uncertainty (k=2) of the error of indication at each test point. The error of indication and its associated uncertainty constitute the calibration result.

6. Acceptance Criteria

The instrument passes calibration where the error of indication at each test point, together with its uncertainty, falls within the tolerance class declared for the instrument (or the client's stated tolerance requirement). Results outside tolerance are reported as a calibration result without a pass/fail statement unless a specific conformity statement has been requested, per ISO/IEC 17025 Section 7.8.6.

Prepared By

Technical Manager

Reviewed By

Quality Manager

Approved By

Managing Director